

SYMPOSIUM REPORT

Endurance exercise-induced and mental fatigue and the brain

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Abstract

In sports and exercise science, fatigue is an elusive concept that has important implications in performance during exercise. It has been described in many ways (tiredness, exhaustion, lethargy or weariness) and describes a physical and/or mental state of being tired and lack of energy. Exercise-induced fatigue can be

defined as an acute impairment of exercise performance, and a distinction has been made between peripheral and central fatigue. Mental fatigue can be defined as a psychobiological state caused by prolonged exertion that has the potential to reduce cognitive performance and exercise performance. Recent studies have given clear indications that brain catecholamines are involved in the onset of fatigue during endurance exercise. Evidence is provided indicating that the noradrenergic neurotransmitter system hastens central fatigue, a finding that coincides with a faster rate of increase in the rating of perceived exertion. Brain neurotransmission is also suggested to play an important role in mental fatigue. Several neurotransmitter systems might be implicated (with the most important role for dopamine and adenosine) in multiple brain regions, such as the prefrontal cortex and the anterior cingulate cortex, and the summation of these alterations might explain the impairment in endurance performance in a mentally fatigued state. Obviously, we have to keep in mind that fatigue is a very complex construct and that, besides brain neurochemistry, several other factors play a role in its onset.

KEYWORDS

brain, fatigue, performance

1 | INTRODUCTION

'Fatigue' is probably one of the most studied phenomena. It is labelled and debated as a physiological destination, a perception or emotion, and an important mechanism to minimize physical injury. It is an experimental concept, a symptom, a risk, a cause and a consequence (Pattyn, Van Cutsem, Dessy, & Mairesse, 2018). This array of facets of fatigue means that it surfaces in exercise physiology, cognitive psychology, human factors and engineering, and medical practice (Pattyn et al., 2018). In sports and exercise science, fatigue is an elusive concept that has important implications in physical and cognitive performance during exercise. It has been described in many ways (tiredness, exhaustion, lethargy or weariness) and describes a physical and/or mental state of being tired and lack of energy.

'Exercise-induced fatigue' can be defined as an acute impairment of exercise performance that includes both an increase in the perceived effort necessary to exert a desired force or power output and the

eventual inability to produce that force or power output (Davis & Bailey, 1997). It is a multidimensional concept, and a distinction has been made between peripheral and central fatigue. Peripheral fatigue involves impairments located in the muscle, whereas central fatigue involves decreased motor output from the primary motor cortex, is associated with modulations at anatomical sites proximal to nerves that innervate skeletal muscle and is referred to as a progressive decline in the ability to activate muscles voluntarily (Tanaka, Ishii, & Watanabe, 2015). Taylor, Amann, Duchateau, Meeusen, and Rice (2016) underlined the importance of brain neurochemistry (i.e. brain neurotransmitters) in this process. Neurotransmitters indeed dictate and create the communication between neurons. Within fatigue research, most emphasis has been given to serotonin and dopamine (Cordeiro et al., 2017), but more recently the role of noradrenaline has also been studied.

Over the last decade, 'mental fatigue' has gained significant attention. Mental fatigue can be defined as a psychobiological state

caused by prolonged exertion that has the potential to reduce cognitive and physical performance (Van Cutsem et al., 2017). Mental fatigue has been shown to exert a negative effect on several aspects of performance (endurance and technical/tactical skills; Pageaux & Lepers, 2018), but not all. Indeed, the shorter and more maximal the task, the lower the impact of the mental fatigue (Van Cutsem et al., 2017). This is explained by the reduced to non-existent cognitive component of such exercise tasks (Martin, Thompson, Keegan, Ball, & Rattray, 2015). Interestingly, the observed decrement in endurance and technical performance was not mediated by an exacerbation of central fatigue or cardiorespiratory factors (Marcora, Staiano, & Manning, 2009). The only rather persistent finding that explains the decrease in performance is a higher than normal perception of effort (Van Cutsem et al., 2017). This 'perception of effort' allows inter-individual differences to be accounted for, in addition to the situational variations in fatigue, and it is applicable to both mental and physical constructs. It integrates motivational and emotional dimensions and overcomes current polemics in various research fields (Pattyn et al., 2018). As with exercise-induced fatigue, brain neurochemistry, and mainly dopamine and adenosine, have been implicated in the onset of mental fatigue and its effects on subsequent physical or cognitive performance.

This symposium report provides an overview of the recent papers linking brain neurotransmission with exercise-induced and/or mental fatigue.

2 | CENTRAL FATIGUE AND THE BRAIN

Contrary to early hypotheses suggesting that an increase in brain serotonin concentration induces fatigue, it appears that serotonin does not act as the key factor. Instead, the central catecholamines dopamine and noradrenaline are thought to play the main roles in central fatigue (Connell, Thompson, Turuwhenua, Szrigh, & Gant, 2017). We refer the readers to previous review articles that have made in-depth contributions regarding their role in the onset of fatigue and in different environmental conditions (Meeusen & Roelands, 2018; Roelands, De Pauw, & Meeusen, 2015).

More recent studies have focused on the role of noradrenaline during and after physical effort. Klass et al. (2012) tested the effects of oral administration of noradrenaline (reboxetine) and dopamine (Ritalin; methylphenidate) reuptake inhibitors (thus increasing brain concentrations of noradrenaline and dopamine, respectively) on acute cycling performance and central fatigue. Their cycling performance data confirmed the results of previous studies (Roelands et al., 2008a, b). Administration of reboxetine significantly decreased time trial performance, whereas the inhibition of dopamine reuptake did not delay the onset of fatigue at 18°C. Interestingly, the reduction in maximal voluntary contraction torque was slightly (although not significantly) greater in the reboxetine conditions compared with the other conditions. The noradrenergic manipulation did not aggravate peripheral fatigue, because there was no difference in the electrically

New Findings

- **What is the topic of this review?**

It provides an overview of the recent papers linking brain neurotransmission with exercise-induced and/or mental fatigue.

- **What advances does it highlight?**

The noradrenergic neurotransmitter system hastens central fatigue during prolonged exercise, a finding that coincides with a faster rate of increase in the rating of perceived exertion. 2) Mental fatigue affects several neurotransmitter systems, with presumably an important role for dopamine and adenosine, in multiple brain regions such as the prefrontal cortex and the anterior cingulate cortex.

induced resting twitch and estimated resting twitch (decreased in all conditions). This led the authors to conclude that the reason for the reduction in performance should be sought centrally. Klass et al. (2012) were able to show the presence of central fatigue through a significantly reduced voluntary activation (tested by transcranial magnetic stimulation) ≤ 30 min after exercise cessation in the reboxetine trial. Further evidence for the central component came from the worsened cognitive performance (increased reaction time on the psychomotor vigilance task) after the reboxetine trial.

Klass, Duchateau, Rabec, Meeusen, and Roelands (2016) progressed on these findings by looking at the neural impairment induced by noradrenaline reuptake inhibition and its effect on the endurance time of a fatiguing task involving intermittent submaximal isometric contractions of the quadriceps muscle repeated to failure. Although this did not elicit large increases in ventilation or cardiac output, the results and conclusions were very similar to those of their previous study on whole-body exercise (Klass et al., 2012). There was a decreased time to exhaustion during the intermittent contractions of the knee extensors after reboxetine administration (-15.6%) compared with the placebo conditions. This reduced endurance time was accompanied by a greater rate of increase in ratings of perceived exertion. No differences were observed in the time course of muscular and corticospinal excitability. The rate of decline in voluntary activation was also greater after reboxetine was administered (Klass et al., 2016). The findings of that study also suggested that a proposed increase in the brain concentration of noradrenaline contributes to the development of central fatigue (decreased output from the motor cortex). Unfortunately, these studies were not able to measure the brain noradrenaline concentrations directly.

In general, we can conclude that brain neurochemistry is involved in the complex regulation of fatigue during endurance exercise, but what that role is remains to be determined, because many mediators play a role. It appears that the noradrenergic neurotransmitter system hastens central fatigue, a finding that coincides with a faster rate of increase in the rating of perceived exertion during exercise.

3 | MENTAL FATIGUE AND THE BRAIN

The precise mechanisms of the onset mental fatigue remain elusive, as does the explanation for the negative impact of mental fatigue on subsequent performance. Until now, the most persistent finding in studies looking at the effects of a mentally fatiguing task on performance is an increase in perception of effort. Several hypotheses have already been put forward (Van Cutsem et al., 2017) to explain how this might work: the afferent feedback model, in which feedback from working muscles and other physiological systems is integrated (Noble & Robertson, 1996); the corollary discharge model, in which neural signals are sent from premotor/motor areas to sensory areas of the brain (Marcora, 2009); and the combined model, using both afferent feedback and corollary discharges to explain the increase in the rating of perceived exertion.

After mentally fatiguing tasks, modifications occur in electrical brain activity patterns (i.e. increased beta band activity in the prefrontal cortex) that are related to increased perceived exertion (Brownsberger, Edwards, Crowther, & Cottrell, 2013). Schiphof et al. (2018) postulated that these alterations in brain activation and the concurrent changes in brain neurotransmitter concentrations mediate between athletes' perceptions (for example of mental fatigue) and their drive to exercise (Schiphof-Godart, Roelands, & Hettinga, 2018). Although at present these are mainly hypotheses and speculations that remain to be confirmed with solid evidence from well-designed scientific studies, there is certainly indirect evidence for a link between mental fatigue and brain neurotransmission. The most frequently suggested, and indirectly studied, neurotransmitters at this stage are dopamine and adenosine, or an interaction of both (Johansson et al., 2014; Pageaux, Lepers, Dietz, & Marcora, 2014). Pageaux and co-workers (Pageaux & Lepers, 2018; Pageaux et al., 2014) hypothesized that prolonged mental exertion could induce accumulation of adenosine in the anterior cingulate cortex, leading to a higher than normal perception of effort during subsequent endurance exercise. Earlier studies have indeed shown that the anterior cingulate cortex is strongly activated during cognitive response inhibition tasks and that this brain area is associated with the perception of effort (Williamson et al., 2002). Additionally, animal studies suggest that neural activity increases extracellular (i.e. synaptic cleft) concentrations of adenosine (Lovatt et al., 2012) and that brain adenosine impairs endurance performance (Davis et al., 2003). In their elaborative review on the role of adenosine, Martin, Meeusen, Thompson, Keegan, and Rattray (2018) progressed on the hypothesis put forward by Pageaux et al. (2014) and proposed that with demanding cognitive activity, extracellular cerebral adenosine would indeed accumulate within active regions of the brain, such as the anterior cingulate cortex. The authors also described that adenosine acts in two ways: by increasing perception of effort during subsequent effortful tasks; and by impairing motivation, or the willingness to exert effort, probably via an interaction with dopamine in the anterior cingulate cortex.

Dopamine has also been linked to mental fatigue. Lorist et al. (2009) suggested that mental fatigue could result in decreased dopamine levels, which might negatively affect effort/reward evaluations,

resulting in the choice of low-cost behavioural alternatives. Moeller, Tomasi, Honorio, Volkow, and Goldstein (2012) stated that the longer time on task, and the mental fatigue that manifests, weaken the cognitive oversight functions of the dorsal anterior cingulate cortex that encompass performance monitoring, cognitive control or signalling the need for increased attentional resources to enhance control. A reduced activation of the dorsal anterior cingulate cortex would then increase the need to recruit additional, compensatory regions to uphold performance (Moeller et al., 2012). One such compensatory region could be the dopaminergic midbrain, the location of the ventral tegmental area and substantia nigra. Moeller et al. (2012) were able to confirm that in healthy subjects, midbrain activity in response to error increased when fatigue was escalating, confirming a role of dopamine in sustaining effort during task performance.

Further evidence for a role of an interaction between adenosine and dopamine in mental fatigue comes from studies that have used caffeine as a countermeasure for mental fatigue. Caffeine acts as an antagonist of the adenosine receptor. Given that adenosine has been associated with an inhibition of the release of excitatory neurotransmitters, such as dopamine, which reduce arousal, spontaneous behaviour and affect (i.e. pleasure) during exercise, caffeine indirectly increases the available amount of dopamine in the brain (Franco-Alvarenga et al., 2019). Although the exact mechanistic questions have not yet been answered, progress has been made. In the study by Azevedo, Silva-Cavalcante, Gualano, Lima-Silva, and Bertuzzi (2016), the participants (low to normal caffeine consumers; >0.5 or <8 mg kg⁻¹ day⁻¹) were able to produce a greater amount of work during mental fatigue and caffeine conditions, as evidenced by an increased duration of exercise tolerance (+14%), with similar rating of perceived exertion values in the different conditions. Van Cutsem, De Pauw, Marcora, Meeusen, and Roelands (2018) administered a caffeine-maltodextrine mouth rinse to participants (low caffeine users, 101 ± 97 mg day⁻¹) to counter the negative effects of mental fatigue. The authors observed a slower increase in self-reported mental fatigue, with an increased ability to maintain cognitive performance (higher response accuracy in the Stroop task) in comparison to the placebo mouth rinse. It was also reported that the caffeine-maltodextrine mouth rinse was able to reverse the mental fatigue-associated electrophysiological changes (i.e. decreasing P2 amplitude) occurring with increasing time on task. Finally, Franco-Alvarenga et al. (2019) confirmed previous findings of performance restoration attributable to caffeine administration to subjects [three non-consumers (≤ 40 mg day⁻¹), five occasional consumers (≤ 250 mg day⁻¹) and four daily consumers (consumption of 250–572 mg day⁻¹)], in a mentally fatigued state. However, the authors were unable to demonstrate a correlation with changes in electrical activation in the prefrontal cortex.

The above findings indicate that mental fatigue and its negative effects on performance are not caused by one particular neurotransmitter. Instead, mental fatigue affects several neurotransmitter systems, presumably with important roles for dopamine and adenosine, in multiple brain regions, such as the prefrontal cortex and the anterior cingulate cortex. The summation of these alterations

might explain (in part) the impairment in endurance performance. This hypothesis requires further research. Recently, it was suggested that changes in brain energy metabolism (i.e. a reduction in brain phosphocreatine concentration) could trigger mental fatigue. Van Cutsem et al. (2020) showed experimentally that 7 days of phosphocreatine supplementation resulted in prolongation of cognitive performance.

4 | CONCLUSION

We can conclude that brain neurochemistry is involved in the complex regulation of fatigue, be it mental, exercise-induced or the combination of both. Several neurotransmitter systems have been implicated in the onset of both exercise-induced and mental fatigue, but the most frequently studied and most plausible candidates are the catecholamines dopamine and noradrenaline, and adenosine. We have to keep in mind that fatigue is a very complex construct and that, besides brain neurochemistry, several other factors play a role in its onset.

COMPETING INTERESTS

None declared.

AUTHOR CONTRIBUTIONS

R.M. contributed to the conception of the work and critically revised the work. J.V.C. contributed to the acquisition of data for the work and the drafting of the work. B.R. contributed to the conception of the work and drafting of the work. All authors approved the final version of the manuscript and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All persons designated as authors qualify for authorship, and all those who qualify for authorship are listed.

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